Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/28/2026
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Spaceflight Effects on Plant-Microbe Interactions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	04/01/2021	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2025 per H. Levine/KSC (Ed., 2/24/25). NOTE: End date changed to 03/31/2025 per NSSC information (Ed., 4/8/24).		
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Bakshi, Arkadipta Ph.D. (University of Wisconsin, Madison) Swanson, Sarah Ph.D. (University of Wisconsin, Madison) Barker, Richard Ph.D. (University of Wisconsin System) Hanson, David Ph.D. (University of New Mexico)		
Grant/Contract No.:	80NSSC21K0577		
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Performance Goal Text:			

Task Description:	This work seeks to address: (1) how spaceflight modulates the interactions between plants and microbes and (2) how well microgravity analogs capture the events elicited by the spaceflight environment. Tomato plants will be grown on orbit in the NASA Vegetable Production System (Veggie) hardware on board the International Space Station (ISS) with and without the beneficial rhizosphere microbe Trichoderma hazianum. A third sample will be of this microbe growing under identical conditions on the ISS but without the plants. These samples will be compared to parallel ground controls at 1 x gravity as well as to samples growing on 1-axis and 3D clinostats. Assays will integrate RNAseq-based transcripomics and ionomics (nutrient uptake and distribution) alongside biochemical measures of photosynthesis and stress. These comparisons will provide measures of both how spaceflight affects the plant, the microbe, and the relationship between these organisms and additionally, how well microgravity analogs can reproduce these kinds of events on the ground. In addition, the omics-level data gathered from this study will be compared to the wealth of spaceflight-related omics data available through the GeneLab data repository. Using an approach of orthologous matrix mapping will allow identification of similar genes between diverse species and so allow for comparisons of, for example, the degree of similarity between patterns of gene expression to be compared between different species. Overall this research will help define how spaceflight may modulate plant, microbial, and plant-microbe responses and help understand whether defined beneficial microbes may provide a countermeasure to the deleterious effects of spaceflight on plants. The work will capitalize on the complementary expertise of two groups: the Gilroy lab team (plant spaceflight, transcriptomics) and the Hanson lab (biochemistry, photosynthesis).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed research seeks to address how spaceflight modulates the interactions between plants and microbes using tomatoes and the beneficial soil microbe Trichoderma harzianum. The microbiome around the plant root is recognized as a crucial element in the productivity and hardiness of plants but the complex interactions and chemical signals that occur between plant and microbe have only recently begun to be dissected. Yet, for example, T. harzianum is used as a commercial biostimulant, being added to the soil to pormote plant growth and vigor. The research in this project will further define the molecular components of the interaction between plant root and this fungus and how spaceflight alters these events. Thus, the work will not only provide insight into how plant-microbe interactions are affected by spaceflight but address whether T. harzianum might be used as a biostimulant to counteract some of the stresses of spaceflight on plant growth. This research will also help provide molecular insight into how these interactions occur on Earth. Such a fuller understanding of plant-microbe interactions, especially of T. harzianum-plant root communications and response will be important steps towards optimizing these beneficial interactions and so increasing plant productivity in both space and on Earth.
Task Progress:	The experiment (Advanced Plant Experiment 10, APEX-10) was launched on the NG-20 Cygnus spacecraft. All on-orbit operations were successfully completed in the Vegetable Production System (Veggie) hardware on the International Space Station (ISS) and all the biology samples were returned to Earth on the CREW 7 vehicle. The biological samples consisted of sterile Petri plates containing nutrient gel medium with either tomato seeds alone, tomato seeds inoculated with spores of the beneficial fungus Trichoderma harzianum, or Petri plates inoculated with fungal spores alone. A two mm pore size polyester mesh was incorporated on top of the nutrient gel before planting of the seeds or inoculating with fungal spores to facilitate crew harvesting of materials for rapid freezing at the end of the experiment. Plants were grown under continuous light for 14 days in the Veggie hardware, with crew photography every other day. The experiment also included gas sampling of the air around the Veggies to allow stable carbon isotope analysis for calculation of photosynthetic and physiological parameters such as plant water use efficiency. On day 14, plates were opened in the Life Sciences Glovebox on the ISS, to ensure fungal containment, and biological materials were harvested, frozen between aluminum blocks conditioned to -160˚C, and stored in the Minus Eighty (Degrees Celsius) Laboratory Freezer for ISS (MELFI) for sample return. On-orbit data collection included imaging of the plants and fungi as they grew and these images revealed that, although treatment with Trichoderma did not alter germination rates (all >80%), Trichoderma-treated plants exhibited inhibition of root growth compared to their untreated controls. Trichoderma-treated plants also had shorter roots in spaceflight when compared to paired ground controls. Cabin air was also sampled in orbit for an analysis of stable carbon isotope ratios of the biological materials. This analysis showed that the 13C:12C isotopic ratio is suppressed in spaceflight. Such a shift is consistent with an alteration in primary carbon metabolism in the plant and potentially reduced water use efficiency and water stress inflight. Despite this shift in isotopic ratio, analysis of carbon to nitrogen content indicated there were no significant changes induced by flight. Biochemical analysis of chlorophyll and carotenoid content revealed that spaceflight did not alter photosynthetic pigment accumulation. However, shoots showed a biochemical signature of stress, with significant accumulation of the stress pigment anthocyanin and malondialdehyde (a biochemical measure of oxidative load). Extraction of nucleic acid and subsequent RNAseq were also used to catalog patterns of gene expression. These changes were significantly different between root and shoot samples, with spaceflight upregulating genetic pathways associated with gas exchange and protein folding in the shoot, along with widespread down-regulation in photosynthetic genes. This is of note as the biochemical measurements outlined above indicated this did not translate into reduced photosynthetic pigment levels (at least as relates to the major tomato photosynthetic pigments of the chlorophylls and carotenoids). Root changes in, for example, genes associated with reproduction, ion transport, and a suite of genes associated with DNA replication showed altered expression. While the extent of the spaceflight-induced transcriptional response was similar between plants with and without Trichoderma in terms of number of altered genes and pathways, there was a Trichoderma-specific spaceflight response in the plants. Thus, many of the patterns of response to spaceflight were maintained in the tomato plants in the presence of Trichoderma, such as suppression of pathways related to photosynthesis, reproduction, and DNA replication; but new patterns did emerge, with increases in genetic networks related to mitochondrial function and induction of heat shock proteins, chaperones that often represent response to oxidative stress. Trichoderma itself exhibited extensive growth in spaceflight but similar levels of growth were also seen in the parallel ground controls, suggesting these responses were not wholly related to a flight-specific element such as microgravity. The flight environment does also include elevated CO2, to ~3000 ppm and the growth chambers for the specimens in APEX-10 were in equilibrium with the ambient air for both the flight and ground control samples, with the ground

	control samples being maintained at cabin-like levels of CO2. Trichoderma maintained under these elevated CO2 levels in ground-based studies altered growth and triggered spore production mimicking the effects seen in the flight experiment, tentatively implying some of the spaceflight effects may relate to the elevated CO2. This observation highlights the multifaceted nature of the environment experienced by organisms living in space. Space-specific factors, such as exposure to microgravity and increased levels of ionizing radiation, are combined with elements imposed by the space vehicle, such as high CO2. Understanding the responses of organisms in spaceflight will require dissecting the effects of these factors alone and in combination.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
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